AIRLOCK

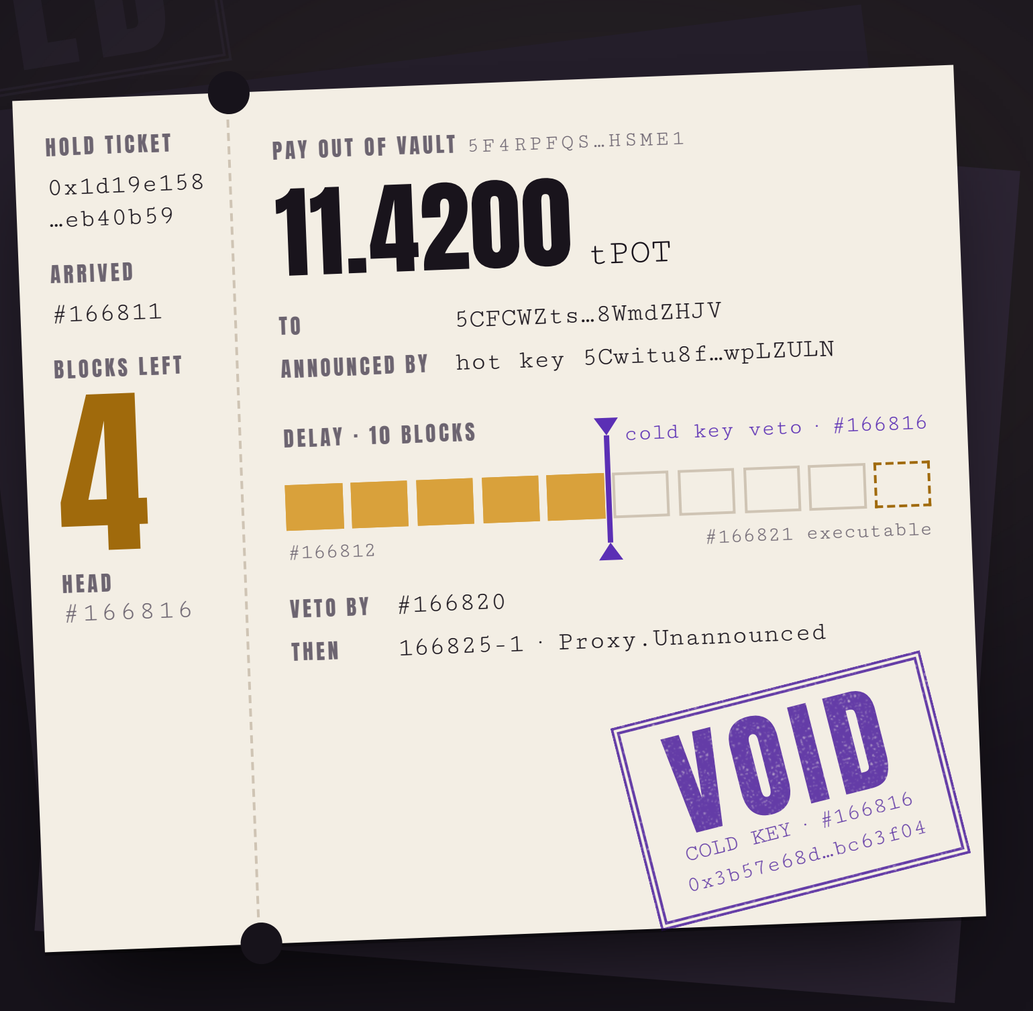
PORTALDOT HACKER HOUSE 2026 · ECOSYSTEM DAPPS

THE HOT KEY THAT HAS TO ASK FIRST.

A Portaldot vault whose hot key can only announce a spend, and whose cold key can veto it.

◆ OWNER · COLD KEY — V3 testnet · 2026-10-24

```
[v3] ok veto by cold #166816 id 166816-1
VETOED 0x1d19e158…eb40b59
```



THE PROBLEM

ONE LEAKED SEED. ONE BLOCK.

POT on a single hot key leaves in one extrinsic once its seed leaks. A multisig stops that, but asks a second signer for every payment.

HOT KEY, NO DELAY

100 UNIT

moved to the thief in one extrinsic.

`control run (bench --ablate) · interim chain, not V3`

TWO-KEY MULTISIG

2 signers

on every payment, for one person.

`README · Problem statement`

THE MECHANISM

ANNOUNCE. WAIT. VETO.

1 · HOT KEY · DELAY 10

ANNOUNCE

It cannot spend directly. It can only publish a call hash.

`Proxy.announce`

2 · 10 BLOCKS ≈ 60 S ON V3

WAIT

The watcher alerts: YOURS, or NOT YOURS with the amount and a veto line.

`event Proxy.Announced`

3 · COLD KEY · DELAY 0

VETO

The thief's execute then fails Proxy.Unannounced.

`Proxy.reject_announcement`

Your own spend: register the intent, announce, wait, execute. No co-signer.

REFUSED ON V3. 4 BLOCKS TO SPARE.

HOLD TICKET
0x1d19e158
…eb40b59

ARRIVED
#166811

BLOCKS LEFT
4

HEAD
#166816

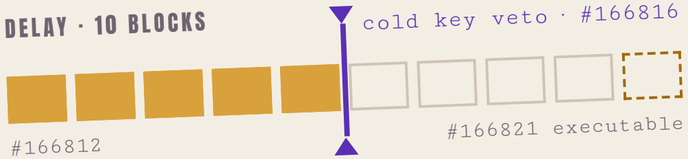
PAY OUT OF VAULT 5F4RPFQS…HSME1

11.4200 tPOT

TO 5CFCWZts…8WmdZHJV

ANNOUNCED BY hot key 5Cwitu8f…wpLZULN

DELAY · 10 BLOCKS



VETO BY #166820

THEN 166825-1 · Proxy.Unannounced

VOID
COLD KEY · #166816
0x3b57e68d…bc63f04

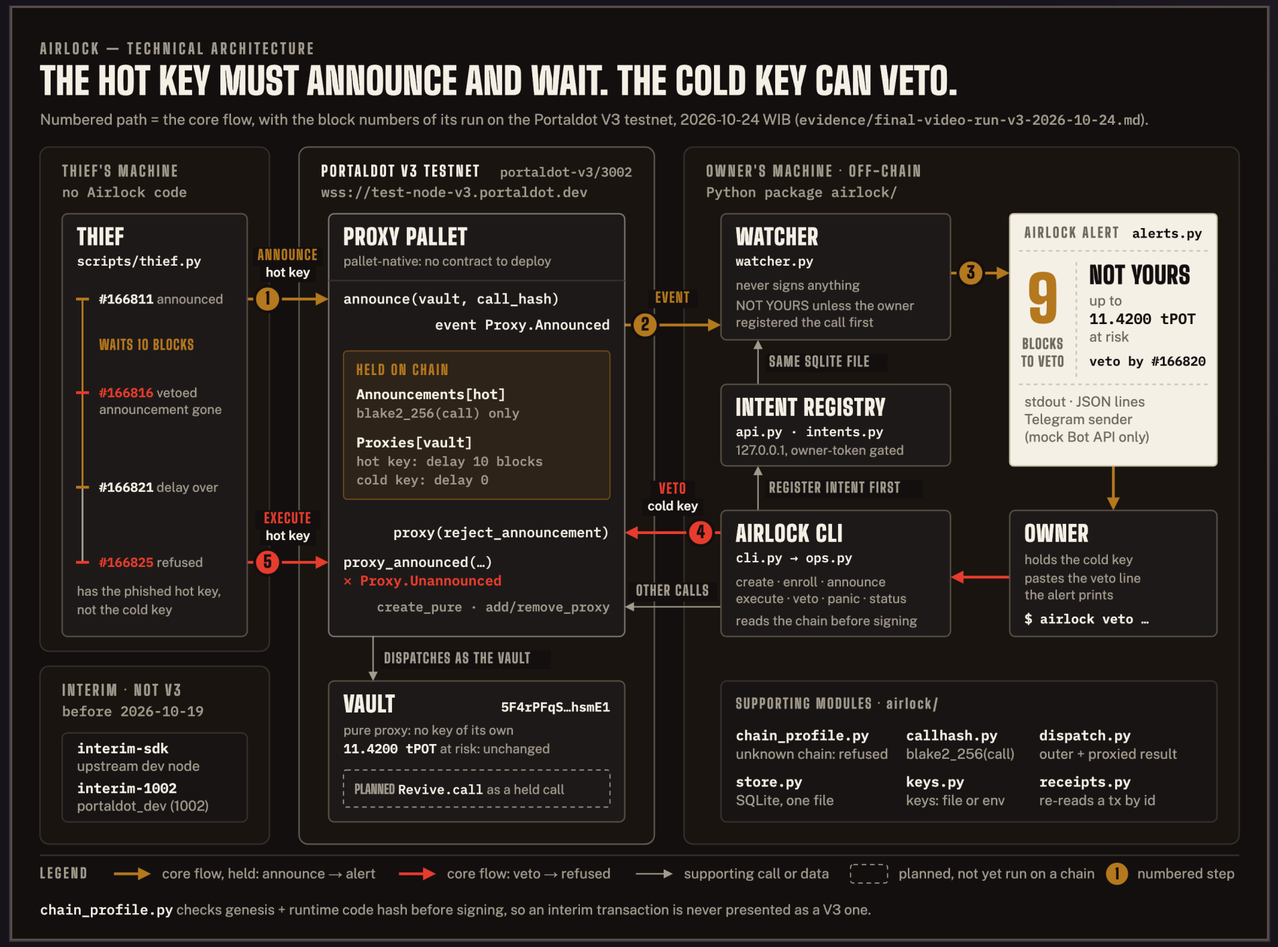
◆ WATCHER · OWNER'S SIDE watcher.cast

NOT YOURS – VETO
Up to 11.4200 tPOT at risk
Veto must land by #166820: 9 blocks (≈ 54 s)
airlock veto --vault 5F4rPFqS… --hash 0x1d19e158…

166811-1	thief announces · ok
166816-1	cold key veto · VETOED
166825-1	execute · Proxy.Unannounced
166831-1	direct call · Proxy.Unannounced

4/4 re-fetched by id · vault total unchanged, #166806 to #166835

THE CHAIN HOLDS THE TICKET. AIRLOCK READS IT.



ON CHAIN

The Proxy pallet holds the delay and the veto. No contract, no custom runtime code.

OWNER'S MACHINE

Watcher, intent registry and CLI. The watcher never signs.

BEFORE SIGNING

`chain_profile.py` checks genesis and runtime code hash.

WHY IT'S NEW

THE PRIMITIVE EXISTED. THE PRODUCT DIDN'T.

Delayed proxies with announce and reject ship in the Proxy pallet. In our search (not a market survey) we found no product built on that path.

SETUP	SIGNERS PER PAYMENT	A LEAKED HOT KEY…	THE OWNER ACTS…
ONE HOT KEY	1	drains the account in one extrinsic	after the money is gone
TWO-KEY MULTISIG	2	cannot spend alone	on every payment
DELAYED PROXY, NO WATCHER	1	can only announce; no alert goes out	only if they notice the hash in time
AIRLOCK	1	can only announce, then wait	only on NOT YOURS: the cold key vetoes

ECOSYSTEM FIT

PORTALDOT'S PROXY PALLET ENFORCES IT.

`Proxy.create_pure` the keyless vault	`Proxy.add_proxy` hot key with delay, cold key with 0	`Proxy.announce` the hot key publishes a call hash
`Proxy.proxy_announced` runs after the delay, if not vetoed	`Proxy.reject_announcement` the veto	`Proxy.remove_proxy` panic: retire the hot key

Security is Portaldot's consensus: a compromised server can mislabel or drop an alert, but cannot move funds or shorten the delay.

Running on Portaldot V3 (portaldot-v3/3002) since 2026-10-19. No contract deployed.
Plan, not built: the hold-ticket flow inside Portaldot Wallet; Revive.call as a held call.

WHOEVER KEEPS A HOT KEY.

POT HOLDERS

A daily hot key for spending; the vault behind it pays out only after a veto window.

GRANTEES

Vested payouts land in a vault, so a leaked laptop key cannot sweep them.

STASHES & TREASURIES

Validator stash funds and small team wallets, one operator key each.

PLAN, NOT BUILT

FIRST USERS

grantees: new grants vested straight into the grantee's vault.

PAID

a hosted watcher with phone alerts; the CLI and watcher stay free.

WALLET

the hold-ticket flow is small enough for Portaldot Wallet to adopt.

STATUS · 2026-10-25

BUILT. STILL OPEN.

ON V3

4/4	2026-10-24 run: veto #166816, execute + direct refused `final-video-run`
7/7	intent registry checks against V3 `890fcfb`
25 · 8	seed extrinsics · call types
44/44	V3 proof rows re-fetched by id `PROOF.md`

INTERIM CHAINS AND OFFLINE

20/20	bench drains vetoed, 0 planck moved · interim, not V3
350	offline tests pass · 192 commits at `937e2f4`

OPEN BEFORE THE FINAL SUBMISSION

- Benchmark on V3 (interim chains only so far)
- Phone alerts (Telegram sender built, no bot)
- Revive.call as a held call (composer only)
- Hosted owner UI (none yet; CLI today)
- Deploy the watcher + registry (local today)

THE ASK: OPEN ANY ID AND CHECK IT

CHECK IT BY EXTRINSIC ID.

PAGE	airlock.edycu.dev
SOURCE	github.com/edycutjong/portaldot-hackathon-2026-airlock-edycu
PROOF LOG	docs/PROOF.md · every row: id, block, tx
FILM	youtu.be/Es1387HDSTw
THE VETO	166816-1 on node-console

AI assistance. Drafted with Claude (Claude Code) and reviewed by the author: most of the code, the tests, the README and this deck. The author's own: the choice of the Proxy-pallet mechanism, the vault layout and the on-chain design decisions (edycu, solo).

SCAN: AIRLOCK.EDYCU.DEV